

Individual Assignment 2

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You should complete given exercises and submit them in a single zip file named as your student ID (e.g. u210023.zip) to eClass before the deadline.

Your completed assignments will not be graded, but will be marked for completion. The maximum mark for completing all assignments in this semester is 10 points. Although it is an individual assignment and you cannot use other student's work in this assignment, you can still use publicly available online resources in this assignment. You must properly reference all resources. Otherwise, it will be considered as plagiarism.

1 Identifying Customer Needs - Toothbrush

You are requested to conduct a customer needs research for a new product which is going to help people in maintaining their oral hygiene. For that you should interview at least 3 people with their personal toothbrush in hand. It can be your friends or family members ¹. Below are the steps you should complete:

1. Observe how your family members or friends use their toothbrush in the morning or evening.
2. After finishing your observation, take some notes on latent needs you have identified so far.
3. Then invite your customer to a short interview. Ensure that your customer holds her toothbrush during the interview. It helps her to feel connected to this simple product during the interview. Ask following questions:
 - What does she like about her toothbrush? Why did she buy this particular model?
 - What does she don't like about her toothbrush?
 - Does she feel anything unpleasant while toothbrushing? Or let's forget about toothbrush for a moment. Any discomfort not related to toothbrush she feels while she is doing her everyday morning rituals? How does she manage this discomfort?
 - What possible improvements could she suggest to make her experience more pleasant?
4. After interview, write down customer statements for your questions.
5. Interpret customer statements into customer needs in the form of a product attribute. Gather at least 10-15 customer needs.
6. Determine the importance of every customer need along with your interviewees. Categorize customer needs into *must-haves*, *performance*² and *delighters*. Take special note on latent needs.
7. Your final output for this task should be a table similar to shown in Figure 1. Use the word **Product** as a neutral term to describe the attributes of your new product.

¹given that they use toothbrush everyday and thus belong to your target market

²the better this feature is, the more satisfied is the customer

No.		Need	Imp.
1	The suspension	reduces vibration to the hands.	3
2	The suspension	allows easy traversal of slow, difficult terrain.	2
3	The suspension	enables high-speed descents on bumpy trails.	5
4	The suspension	allows sensitivity adjustment.	3
5	The suspension	preserves the steering characteristics of the bike.	4
6	The suspension	remains rigid during hard cornering.	4
7	The suspension	is lightweight.	4
8	The suspension	provides stiff mounting points for the brakes.	2
9	The suspension	fits a wide variety of bikes, wheels, and tires.	5
10	The suspension	is easy to install.	1
11	The suspension	works with fenders.	1
12	The suspension	instills pride.	5
13	The suspension	is affordable for an amateur enthusiast.	5
14	The suspension	is not contaminated by water.	5
15	The suspension	is not contaminated by grunge.	5
16	The suspension	can be easily accessed for maintenance.	3
17	The suspension	allows easy replacement of worn parts.	1
18	The suspension	can be maintained with readily available tools.	3
19	The suspension	lasts a long time.	5
20	The suspension	is safe in a crash.	5

Figure 1: Customer Needs example

2 Determining Product Specifications - Toothbrush

You should use your collected customer needs to determine specifications for your new product. This will involve investigating technical parameters that are commonly used for measuring the quality of a toothbrush and creating your own metrics that can be objectively measured. Below are the steps you should complete:

1. Convert each customer need into measurable units. Research internet or use your engineering creativity to come up with reliable metrics that can measure not only technical parameters of the toothbrush, but also the human experience while using the toothbrush. For the latter, you may involve metrics that would require experiments with human subjects. Try to identify at least 15 engineering metrics that will address your customer needs.
2. Map each engineering metric to a customer need in the form of a QFD matrix shown in Figure 2. Determine the strength of relationship between customer needs and engineering metrics using fixed grades *strong* (9 points), *medium* (4 points) and *weak* (1 point).

		Performance measure			Size of range		Technical details		
	Customer importance	Meet standard	Harness weight	Webbing strength	No. of colors	No. of sizes	Padding thickness	No. of buckles	No. of gear loops
Easy to put on	2					○		●	
Comfortable when hanging	5					○	●	○	
Fits over different clothes	1					○	○	●	
Accessible gear loops	3								●
Does not restrict movement	5		○			○	●	○	
Light weight	3		●	○			○	△	△
Safe	5	●	○	●					
Attractive	2		△		●		△	△	
Absolute importance		45	67	57	18	52	108	72	30

Figure 2: QFD example

3. Calculate the absolute importance for each engineering metric. Identify top 3 metrics that will make the most impact for customers.
4. Research the market for competitive products, find at least 3 products that can be considered as your competitor. Measure them using engineering metrics you have identified above, or use your best educated guess.
5. Use your market research to identify ideal and acceptable targets for your engineering metrics.
6. Your final output for this should be a table similar to shown in Figure 3.

	Metric	Units	Marginal Value	Ideal Value
1	Attenuation from dropout to handlebar at 10hz	dB	>10	>15
2	Spring pre-load	N	480 - 800	650 - 700
3	Maximum value from the Monster	g	<3.5	<3.2
4	Minimum descent time on test track	s	<13.0	<11.0
5	Damping coefficient adjustment range	N-s/m	0	>200
6	Maximum travel (26in wheel)	mm	33 - 50	45
7	Rake offset	mm	37 - 45	38
8	Lateral stiffness at the tip	kN/m	>65	>130
9	Total mass	kg	<1.4	<1.1
10	Lateral stiffness at brake pivots	kN/m	>325	>650
11	Headset sizes	in	1.000 1.125	1.000 1.125 1.250
12	Steertube length	mm	150 170 190 210	150 170 190 210 230
13	Wheel sizes	list	26in	26in 700c
14	Maximum tire width	in	>1.5	>1.75
15	Time to assemble to frame	s	<60	<35
16	Fender compatibility	list	none	all
17	Instills pride	subj	>3	>5
18	Unit manufacturing cost	US\$	<85	<65
19	Time in spray chamber w/o water entry	s	>2300	>3600
20	Cycles in mud chamber w/o contamination	k-cycles	>15	>35
21	Time to disassemble/assemble for maintenance	s	<300	<160
22	Special tools required for maintenance	list	hex	hex
23	UV test duration to degrade rubber parts	hours	>250	>450
24	Monster cycles to failure	cycles	>300k	>500k
25	Japan Industrial Standards test	binary	pass	pass
26	Bending strength (frontal loading)	MN	>70	>100

Figure 3: Product Specification